

Abandoned exoskeleton  
remains hanging  
from foliage

## Change of skin

Some insects change shape gradually, each time they shed their "exoskeleton" body case. Their wings develop from small buds, and so only the adults can fly. This bush cricket is finishing its final molt. Its old exoskeleton has split open and the adult insect climbs out with fully formed wings, ready to breed. This growing process is called gradual, or incomplete, metamorphosis, because the changes are small. Dragonflies and cockroaches also develop in this way.



## Young adrift

Metamorphosis is very common among sea animals and many invertebrates have young, or larvae, that drift with the currents. Over the weeks or months, the larvae slowly grow and change shape. They often settle on the shore or seabed before becoming adults.



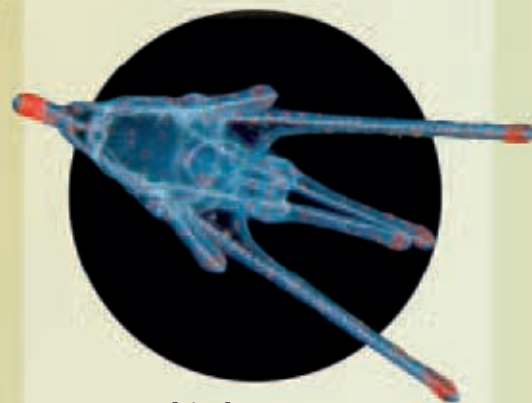
### Barnacle larva

This newly hatched barnacle larva has a single eye. It swims and feeds by using its feathery antennae.



### Crab larva

Halfway to adulthood, this crab larva has large eyes and strong legs. It lives and feeds on the seabed.



### Sea urchin larva

Long arms, reinforced by hard mineral spikes, help this sea urchin larva to drift near the surface of the ocean.

Eye not yet  
fully formed

12-day-old embryo

14-day-old  
tadpole  
leaves jelly

## From water to land

Great crested newts lay their eggs in ponds. Their newly hatched tadpoles are tiny and almost transparent, with feathery gills and no trace of any legs. During the following four months, the tadpoles grow up fast and their bodies transform. Their gills shrink and they develop lungs and spindly legs. The fully grown newts can breathe with their lungs as well as through their skin, ready for life on land and in the water. Several years then pass before the adult newts are old enough to breed.

Egg in jellylike case

Spindly leg not yet  
ready for walking  
on land

External gills allow  
breathing in water

28-day-old tadpole  
grows legs

Male adult newt returns  
to pond to breed

Five toes on hind leg